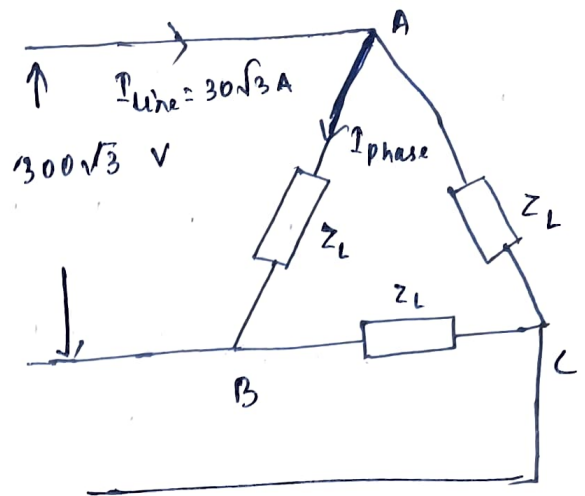


①

In Δ -connected system,

Line voltage = phase voltage

$$|\text{phase current}| = \frac{|\text{Line current}|}{\sqrt{3}}$$



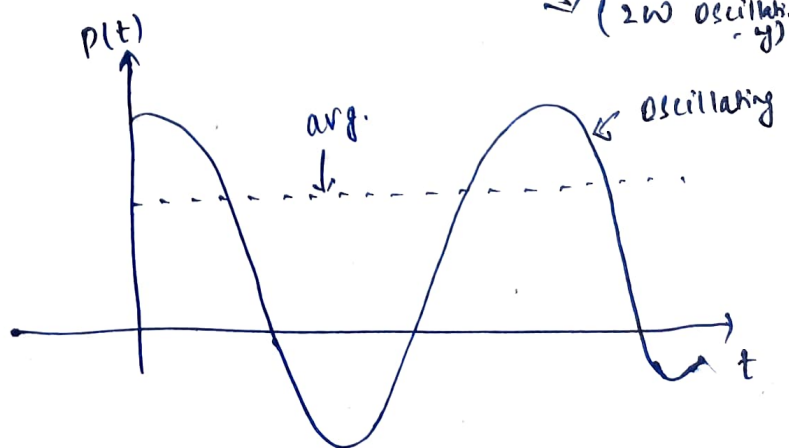
so, $|I_{\text{phase}}|_A = 30 \text{ A}$

$$|Z_{\text{phase}}| = \frac{|V_{\text{phase}}|}{|I_{\text{phase}}|} = 10\sqrt{3} \Omega \quad \underline{\underline{\text{Ans.}}}$$

2. In single phase: $p(t) = v(t) \cdot i(t)$

$$= V_m I_m \cos(\omega t + \theta_v) \cos(\omega t + \theta_i)$$

so,
$$p(t) = \underbrace{0.5 V_m I_m \cos(\theta_v - \theta_i)}_{\text{Avg.}} + \underbrace{0.5 V_m I_m \cos(2\omega t + \theta_v + \theta_i)}_{(2\omega \text{ oscillating})}$$



In 3- ϕ system :-

$$P = P_a + P_b + P_c = V_{AN} i_a + V_{BN} i_b + V_{CN} i_c$$

$$P = 2 V_p I_p \left[\cos \omega t \cos(\omega t - \theta) + \cos(\omega t - 120) \cos(\omega t - 120 - \theta) + \cos(\omega t + 120) \cos(\omega t + 120 - \theta) \right]$$

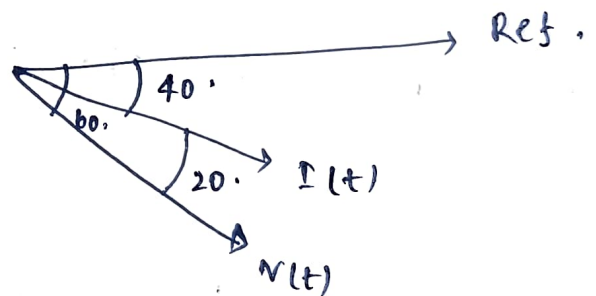
$$\boxed{P = 3 V_p I_p \cos \theta} \quad \leftarrow \text{constant}$$

③ Given $V(t) = 10 \cos(4t - 60^\circ) \text{ V}$

$$I(t) = 4 \sin(4t + 50^\circ) \text{ A}$$

$I(t)$ can be written as $4 \cos(4t - 40^\circ) \text{ A}$ also

So, power factor = $\cos 20^\circ$ (lead)
 $= 0.94$ (lead)



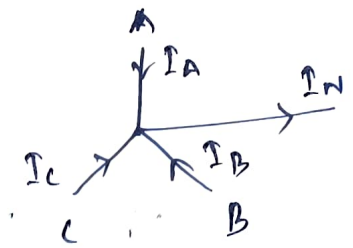
④ $\Rightarrow P = VI$

for constant $P \Rightarrow V \propto \frac{1}{I}$

if V increased, I reduced and hence $I^2 R$ gets lower

$\Rightarrow$ Transformer is used for voltage transformation
(either step up/down) for AC system.

- (5) In a balanced γ system
 $\Rightarrow I_N = I_A + I_B + I_C = 0$



$\Rightarrow$ voltage transformation is done in AC system through transformers in AC system.

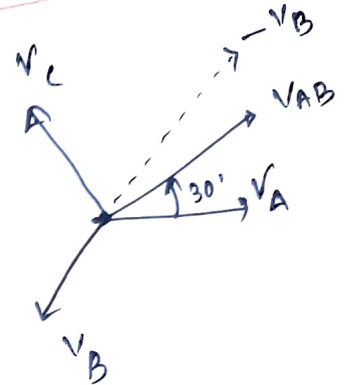
$\Rightarrow$ Also, AC-3 phase carry more power.

Marks to all

- (6) In star connection (3- ϕ)

Line voltage leads phase voltage by 30° .

Line current = phase current.



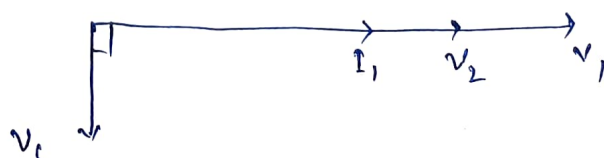
(7) $X_C = \frac{-j}{2\pi f C} \Rightarrow X_C \propto \frac{1}{f}$

$X_L = j2\pi L f \Rightarrow X_L \propto f$

- (8) When $X_L = X_C \Rightarrow$ both of them become short circuit together

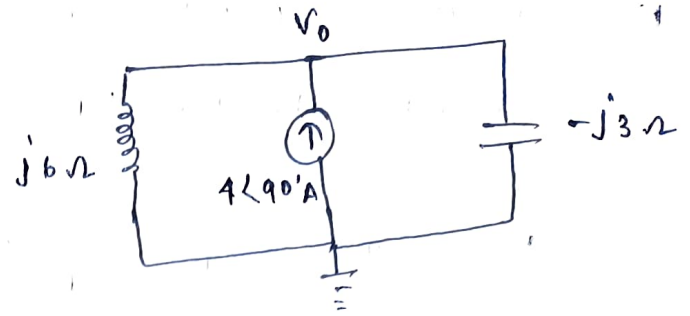
So, circuit becomes resistive. So, I will be in phase with V_1 . V_2 will be voltage across resistor,

& V_C will lag 90° by I .



9

Apply nodal analysis at V_0



$$-4\angle 90^\circ + \frac{V_0}{6\angle 90^\circ} + \frac{V_0}{3\angle -90^\circ} = 0$$

$$V_0 = 2.4 \text{ V}$$

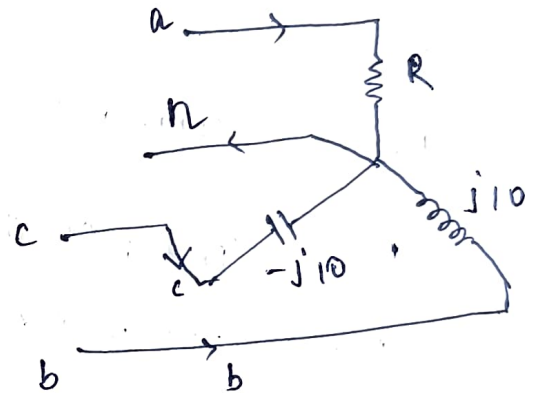
10

$$I_n = I_a + I_b + I_c$$

$$I_n = 0 \text{ (A/q)}$$

$$\text{so, } I_a + I_b + I_c = 0$$

$$\frac{V_{an}}{R} + \frac{V_{bn}}{j10} + \frac{V_{cn}}{-j10} = 0$$



Putting value of V_{an} , V_{bn} & V_{cn} & solve

$$R = 5.77 \Omega$$

11

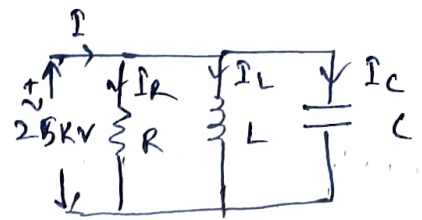
$R = 2 \text{ k}\Omega$, $L = 40 \text{ mH}$, $C = 1 \mu\text{F}$ (series resonant ckt)

$$Q = \frac{1}{R} \sqrt{L/C} = 0.1$$

(12)

Under resonance, L & C

together become open.



$$\text{so, } I = I_R = \frac{V}{R} = \frac{25 \text{ kV}}{5 \text{ k}\Omega} = 5 \text{ A}$$

$$Q = R \sqrt{C/L} = 433.01$$

$$|I_L| = Q I = 5 \times 433.01 = 2.165 \text{ kA}$$

(13)

for Δ connected load, $V_{\text{line}} = V_{\text{ph}}$

$$P = 3 V_{\text{ph}} I_{\text{ph}} \cos \phi$$

$$P = 3 \times 230 \times 15 \times \cos 30^\circ$$

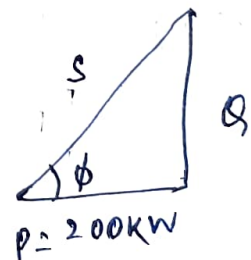
$$P = 8.96 \text{ kW}$$

(14)

when pf is 0.5 lag

$$\phi = \cos^{-1}(0.5) = 60^\circ$$

$$S = \frac{P}{\cos \phi} = 400 \text{ kVA}$$

when pf is unity i.e. $Q = 0$

$$\text{so, } P = S \text{ (under same kVA rating)}$$

$$\text{so, } 200 \text{ kW extra power}$$

(15)

Given $V = 50 \sin(377t + 30^\circ) \text{ V}$

$$V_{\text{peak}} = 50 \text{ V}$$

$$V_{\text{rms}} = \frac{V_{\text{peak}}}{\sqrt{2}} = \frac{50}{\sqrt{2}} \text{ V} = 35.36 \text{ V}$$

So,

$$V_{\text{rms}} = 35.36 \text{ V}$$